Higher National Diploma in (Electrical) Engineering

Fourth Year, First Semester Examination – 2019

EE 3211 - Electrical Machine Control and Protection

Instructions for Candidates:
Answer any five questions.
All questions carry 20 marks.
Calculators are allowed.

No of pages : 04
No of questions : 06
Time: Three (03) hours

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01).

The speed of a 10 hp, 230 V, 1200 rev/min separately excited DC motor is controlled by a single-phase fully-controlled thyristor converter. The rated motor current is 38 A, and the armature resistance is Ra = 0.3Ω. The ac supply is 260 V. The motor voltage constant K = 0.182 V/rev/min. Assume that sufficient inductance is present in the armature circuit to make the armature current continuously and ripple free. Assume firing angle of the converter $\alpha=30^0$ and rated motor current.

i) Draw the circuit diagram and output voltage waveform (5 Marks)

ii) Find the motor torque (5 Marks)

iii) Operating speed of the motor (5 Marks)

iv) Supply power factor (5 Marks)

[TOTAL MARKS 20]

02).

i) 1) Name the types of Braking of Electrical Machines

(3 marks)

2) Explain the Braking types **(5 marks)**

ii) Draw and explain Thermal Overload Relay **(4 marks)**

iii) Mention the applications and advantages/ disadvantages of the above Thermal Overload Relay

(4 marks)

iv) Draw and explain operation of HRC fuse and write its advantages

(4 marks)

[TOTAL MARKS 20]

03).

i) Mention methods of Speed control of Induction Motor

(4 marks)

ii) Briefly explain three of them **(6 marks)**

iii) A 230 V, 10 hp d.c. shunt motor delivers power to a load at 1200 r/min. The armature current drawn by the motor is 200 A. The armature circuit resistance of the motor is 0.2 Ω and the field resistance is 115Ω. If the rotational losses are 500W, what is the value of the load torque?

(Note that 1 hp = 746 W) **(10 marks)**

[TOTAL MARKS 20]

04).

i). 1) Mention the types of Soft Starters.

(3 marks)

2) Explain three types of Soft Starters **(6 marks)**

ii). find which type of motor is suitable for various service **(3 marks)**

	Type of industry equipment	Type of motor
I	Conveyors	.....
II	Electric train	.....
III	Pumps	.....

iii). A series-connected DC motor has an armature resistance of 0.5Ω and field winding resistance of 1.5Ω. In driving a certain load at 1200 rpm, the current drawn by the motor is 20A from a voltage source of VT = 220V. The rotational loss is 150W. Find the output power and efficiency.

(8 marks)

[TOTAL MARKS 20]

05)

i) A separately excited DC motor runs at 500 rev/min when operated from the rated supply voltage of 230V. The load torque is proportional to the square of the speed and the armature current drawn at this speed is 32 A. The armature circuit resistance is 0.7Ω. Calculate the required armature voltage if the speed is to be reduced by half. **(10 Marks)**

ii) The synchronous reactance per phase of a 3 phase star connected 6600V synchronous motor is 20 Ω. For a certain load input, the input is 915Kw at normal voltage and the induced line emf is 8942V. Neglecting resistance and determine the following,

1) Line current
2) Power factor

(10 marks)

[TOTAL MARKS 20]

06)

i). 1) Write protection types of generators (6 marks)

2) explain two of them (4 marks)

ii). A 480V, 50hp, three phase induction motor is drawing 60A at 0.85 pf lagging. The stator copper losses are 2kW and the rotor copper losses are 700W. The friction loss is 600W and the Core losses are 1800W. Find the following

1) The air gap power
2) The converted power
3) The output power
4) The efficiency of the motor (10 marks)

[TOTAL MARKS 20]